

Table 1: Pass-sum execution time (s, median per iteration; sum of pass medians, not full executable wall time) and speedup split by pass type. When present, the OctTree row includes ColorOctree passes; a separate ColorOctree row reports only those passes. ADT fields = total fields in the selected benchmark ADT (prefer parsed ADT definition; fall back to `@BENCH adt_fields`). SoA bufs = 1 tag buffer plus one buffer for each scalar field, plus one buffer for each non-self packed field annotated Linear, plus recursively counted buffers for each non-self packed field annotated Factored; self-recursive fields add no new buffers. Speedup $>1\times$ means the denominator is faster; **bold** marks $>1.1\times$.

Program	ADT	SoA	End-to-end			Fold passes			Map passes		
	fields	bufs	Am (s)	Sm (s)	Am/Sm	Am (s)	Sm (s)	Am/Sm	Am (s)	Sm (s)	Am/Sm
OctTree	16	9	0.6325	0.3748	1.69 $\times$	0.3845	0.3236	1.19 $\times$	0.2479	0.0512	4.84 $\times$
Compiler	9	8	0.2846	0.0611	4.66 $\times$	0.1312	0.0394	3.33 $\times$	0.1534	0.0216	7.09 $\times$
DBQuery	15	13	0.2542	0.1116	2.28 $\times$	0.1261	0.0877	1.44 $\times$	0.1281	0.0239	5.37 $\times$
DecisionTree	7	6	4.066	2.951	1.38 $\times$	4.066	2.951	1.38 $\times$	–	–	–
DomTree	15	14	0.2105	0.1161	1.81 $\times$	0.1540	0.0589	2.62 $\times$	0.0565	0.0572	0.99 $\times$
KDTree	17	16	0.1789	0.1400	1.28 $\times$	0.1789	0.1400	1.28 $\times$	–	–	–
LinearListReduction	11	11	0.0331	0.0050	6.61 $\times$	0.0331	0.0050	6.61 $\times$	–	–	–
reduceNestedList	3	3	0.0849	0.0073	11.57 $\times$	0.0849	0.0073	11.57 $\times$	–	–	–
List	2	2	0.3709	0.2909	1.28 $\times$	0.1412	0.1099	1.28 $\times$	0.2297	0.1810	1.27 $\times$
MonoTree	3	2	0.0465	0.0379	1.23 $\times$	0.0173	0.0221	0.78 $\times$	0.0292	0.0158	1.85 $\times$
ObjectGraph	5	4	0.2121	0.1428	1.49 $\times$	0.0968	0.0805	1.20 $\times$	0.1153	0.0623	1.85 $\times$
PiecewiseFunctions	8	7	0.3515	0.1388	2.53 $\times$	0.1467	0.0922	1.59 $\times$	0.2048	0.0466	4.40 $\times$
TernaryTree	5	3	0.0809	0.0697	1.16 $\times$	0.0285	0.0289	0.98 $\times$	0.0524	0.0408	1.29 $\times$
Trie	10	9	0.1992	0.0707	2.82 $\times$	0.0683	0.0315	2.17 $\times$	0.1309	0.0392	3.34 $\times$
ColorOctree	25	18	0.0953	0.0734	1.30 $\times$	0.0953	0.0734	1.30 $\times$	–	–	–

Table 2: Per-pass performance for **OctTree**, OctTree SoA uses 9 buffers; ColorOctree SoA uses 18 buffers. Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	AoS med $\pm$ err	SoA med $\pm$ err	Speedup
barnesHutPotential	F	11/16	31%	0.0430 $\pm$ 0.00002	0.0392 $\pm$ 0.00101	1.10 $\times$
countActive	F	10/16	38%	0.0372 $\pm$ 0.00002	0.0289$\pm$0.00001	1.29 $\times$
countParticles	F	8/16	50%	0.0364 $\pm$ 0.00110	0.0274$\pm$0.00002	1.32 $\times$
fmmPotential	F	12/16	25%	0.0979 $\pm$ 0.00002	0.0955 $\pm$ 0.00019	1.03 $\times$
sumEnergy	F	12/16	25%	0.0382 $\pm$ 0.00001	0.0321$\pm$0.00001	1.19 $\times$
sumMass	F	10/16	38%	0.0364 $\pm$ 0.00002	0.0272$\pm$0.00002	1.34 $\times$
clearFlags	M	15/16	6%	0.1234 $\pm$ 0.00107	0.0066$\pm$0.00001	18.68 $\times$
scaleEnergy	M	16/16	0%	0.1245 $\pm$ 0.00032	0.0446$\pm$0.00080	2.79 $\times$
paletteEntriesQuantized	F	13/25	48%	0.0459 $\pm$ 0.00086	0.0302$\pm$0.00007	1.52 $\times$
quantizationErrorProxy	F	10/25	60%	0.0494 $\pm$ 0.00003	0.0431$\pm$0.00002	1.15 $\times$
Total				0.6325	0.3748	1.69 $\times$
Geomean						1.76 $\times$

Table 3: Per-pass performance for **Compiler**, ADT has 9 fields, SoA uses 8 buffers. Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	AoS med $\pm$ err	SoA med $\pm$ err	Speedup
blockCountPass	F	2/9	78%	0.0126 $\pm$ 0.00001	0.0019$\pm$0.00000	6.56 $\times$
branchStatsPass	F	2/9	78%	0.0121 $\pm$ 0.00034	0.0036$\pm$0.00000	3.40 $\times$
castInstCountPass	F	3/9	67%	0.0190 $\pm$ 0.00002	0.0013$\pm$0.00000	14.76 $\times$
has cycle	F	4/9	56%	0.0247 $\pm$ 0.00105	0.0175$\pm$0.00016	1.41 $\times$
instCountPass	F	2/9	78%	0.0126 $\pm$ 0.00025	0.0019$\pm$0.00000	6.56 $\times$
latencyModelPass	F	3/9	67%	0.0125 $\pm$ 0.00001	0.0030$\pm$0.00000	4.19 $\times$
memoryOpStatsPass	F	3/9	67%	0.0122 $\pm$ 0.00001	0.0038$\pm$0.00000	3.20 $\times$
throughputModelPass	F	3/9	67%	0.0125 $\pm$ 0.00001	0.0030$\pm$0.00000	4.20 $\times$
verifyIR	F	9/9	0%	0.0129 $\pm$ 0.00047	0.0034$\pm$0.00000	3.75 $\times$
stripSideEffectsPass	M	7/9	22%	0.0775 $\pm$ 0.00019	0.0104$\pm$0.00001	7.43 $\times$
targetReturnPass	M	9/9	0%	0.0760 $\pm$ 0.00017	0.0112$\pm$0.00003	6.78 $\times$
Total				0.2846	0.0611	4.66 $\times$
Geomean						4.82 $\times$

Table 4: Per-pass performance for **DBQuery**, ADT has 15 fields, SoA uses 13 buffers. Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	AoS med $\pm$ err	SoA med $\pm$ err	Speedup
countJoins	F	3/15	80%	0.0196 $\pm$ 0.00001	0.0112$\pm$0.00001	1.75 $\times$
filterSelectivitySkew	F	5/15	67%	0.0196 $\pm$ 0.00001	0.0123$\pm$0.00001	1.60 $\times$
hashJoinPressure	F	5/15	67%	0.0211 $\pm$ 0.00010	0.0126$\pm$0.00002	1.67 $\times$
sumCost	F	6/15	60%	0.0196 $\pm$ 0.00001	0.0148$\pm$0.00002	1.32 $\times$
sumMemory	F	6/15	60%	0.0195 $\pm$ 0.00100	0.0147$\pm$0.00001	1.32 $\times$
sumRows	F	6/15	60%	0.0266 $\pm$ 0.00002	0.0220$\pm$0.00082	1.21 $\times$
clearQueryFlags	M	14/15	7%	0.0640 $\pm$ 0.00084	0.0085$\pm$0.00001	7.49 $\times$
scaleCosts	M	15/15	0%	0.0641 $\pm$ 0.00015	0.0153$\pm$0.00004	4.18 $\times$
Total				0.2542	0.1116	2.28 $\times$
Geomean						2.05 $\times$

Table 5: Per-pass performance for **DecisionTree**, ADT has 7 fields, SoA uses 6 buffers. Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	AoS med $\pm$ err	SoA med $\pm$ err	Speedup
classify Batch	F	5/7	29%	3.488 $\pm$ 0.0032	2.604$\pm$0.0086	1.34 $\times$
classify Depth	F	4/7	43%	0.0500 $\pm$ 0.00001	0.0369$\pm$0.00004	1.35 $\times$
classify tree	F	5/7	29%	0.0500 $\pm$ 0.00015	0.0369$\pm$0.00005	1.35 $\times$
countFeatureUses	F	3/7	57%	0.0534 $\pm$ 0.00014	0.0323$\pm$0.00002	1.66 $\times$
countLeaves	F	2/7	71%	0.0510 $\pm$ 0.00076	0.0242$\pm$0.00001	2.10 $\times$
countNodes	F	2/7	71%	0.0510 $\pm$ 0.00085	0.0248$\pm$0.00078	2.05 $\times$
countSmallLeaves	F	3/7	57%	0.0543 $\pm$ 0.00001	0.0327$\pm$0.00002	1.66 $\times$
inferenceCost	F	2/7	71%	0.0538 $\pm$ 0.00002	0.0305$\pm$0.00001	1.76 $\times$
sumImpurity	F	3/7	57%	0.0534 $\pm$ 0.00001	0.0329$\pm$0.00002	1.62 $\times$
sumPathLengths	F	3/7	57%	0.0558 $\pm$ 0.00001	0.0340$\pm$0.00040	1.64 $\times$
sumSamples	F	3/7	57%	0.0514 $\pm$ 0.00002	0.0321$\pm$0.00003	1.60 $\times$
treeDepth	F	2/7	71%	0.0540 $\pm$ 0.00002	0.0300$\pm$0.00002	1.80 $\times$
Total				4.066	2.951	1.38 $\times$
Geomean						1.65 $\times$

Table 6: Per-pass performance for **DomTree**, ADT has 15 fields, SoA uses 14 buffers. Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	AoS med $\pm$ err	SoA med $\pm$ err	Speedup
count styled	F	3/15	80%	0.0383 $\pm$ 0.00061	0.0115$\pm$0.00002	3.34$\times$
find max Bottom	F	5/15	67%	0.0381 $\pm$ 0.00002	0.0169$\pm$0.00007	2.25$\times$
SumArea	F	6/15	60%	0.0396 $\pm$ 0.00002	0.0191$\pm$0.00002	2.08$\times$
sumTextWidth	F	3/15	80%	0.0381 $\pm$ 0.00001	0.0115$\pm$0.00060	3.33$\times$
computeWidths	M	14/15	7%	0.0302$\pm$0.00006	0.0458 $\pm$ 0.00010	0.66 $\times$
scaleLayout	M	15/15	0%	0.0263 $\pm$ 0.00005	0.0115$\pm$0.00001	2.29$\times$
Total				0.2105	0.1161	1.81$\times$
Geomean						2.07$\times$

Table 7: Per-pass performance for **KDTree**, ADT has 17 fields, SoA uses 16 buffers. Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	AoS med $\pm$ err	SoA med $\pm$ err	Speedup
countInRange tight_box	F	13/17	24%	0.0279 $\pm$ 0.00001	0.0200$\pm$0.00001	1.39$\times$
Find nearest Neighbour	F	13/17	24%	0.0278 $\pm$ 0.00015	0.0213$\pm$0.00012	1.30$\times$
photonMappingBenchmark	F	12/17	29%	0.0422 $\pm$ 0.00001	0.0413 $\pm$ 0.00002	1.02 $\times$
pointCloudNeighborhood	F	11/17	35%	0.0249 $\pm$ 0.00001	0.0165$\pm$0.00001	1.51$\times$
sumMassInRange	F	14/17	18%	0.0274 $\pm$ 0.00001	0.0202$\pm$0.00001	1.36$\times$
twoPointCorrelation bin_8_16	F	11/17	35%	0.0287 $\pm$ 0.00002	0.0208$\pm$0.00001	1.38$\times$
Total				0.1789	0.1400	1.28$\times$
Geomean						1.32$\times$

Table 8: Per-pass performance for **LinearListReduction**, ADT has 11 fields, SoA uses 11 buffers. Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	AoS med $\pm$ err	SoA med $\pm$ err	Speedup
reduction	F	2/11	82%	0.0331 $\pm$ 0.00001	0.0050$\pm$0.00000	6.61$\times$
Total				0.0331	0.0050	6.61$\times$
Geomean						6.61$\times$

Table 9: Per-pass performance for **reduceNestedList**, ADT has 3 fields, SoA uses 3 buffers. Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	AoS med $\pm$ err	SoA med $\pm$ err	Speedup
reduction	F	2/3	33%	0.0849 $\pm$ 0.00004	0.0073$\pm$0.00001	11.57$\times$
Total				0.0849	0.0073	11.57$\times$
Geomean						11.57$\times$

Table 10: Per-pass performance for **List**, ADT has 2 fields, SoA uses 2 buffers. Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	AoS med $\pm$ err	SoA med $\pm$ err	Speedup
length List	F	1/2	50%	0.0460 $\pm$ 0.00030	0.0215$\pm$0.00001	2.14$\times$
sumList	F	2/2	0%	0.0479 $\pm$ 0.00001	0.0434$\pm$0.00014	1.10$\times$
sumList tail recursive	F	2/2	0%	0.0473 $\pm$ 0.00001	0.0450 $\pm$ 0.00001	1.05 $\times$
add1 List	M	2/2	0%	0.2297 $\pm$ 0.00362	0.1810$\pm$0.00032	1.27$\times$
Total				0.3709	0.2909	1.28$\times$
Geomean						1.33$\times$

Table 11: Per-pass performance for **MonoTree**, ADT has 3 fields, SoA uses 2 buffers. Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	AoS med $\pm$ err	SoA med $\pm$ err	Speedup
sumTree	F	3/3	0%	0.0089$\pm$0.00001	0.0112 $\pm$ 0.00002	0.80 $\times$
sumTree TailRec	F	3/3	0%	0.0084$\pm$0.00000	0.0109 $\pm$ 0.00007	0.77 $\times$
add1Tree	M	3/3	0%	0.0292 $\pm$ 0.00095	0.0158$\pm$0.00005	1.85$\times$
Total				0.0465	0.0379	1.23$\times$
Geomean						1.04 $\times$

Table 12: Per-pass performance for **ObjectGraph**, ADT has 5 fields, SoA uses 4 buffers. Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	AoS med $\pm$ err	SoA med $\pm$ err	Speedup
countLargeItems	F	3/5	40%	0.0135 $\pm$ 0.00001	0.0104$\pm$0.00002	1.30$\times$
countMarked	F	3/5	40%	0.0134 $\pm$ 0.00001	0.0110$\pm$0.00066	1.22$\times$
countSurvivors	F	4/5	20%	0.0144 $\pm$ 0.00001	0.0131 $\pm$ 0.00001	1.10 $\times$
deadBytes	F	4/5	20%	0.0148 $\pm$ 0.00001	0.0126$\pm$0.00001	1.17$\times$
liveBytes	F	4/5	20%	0.0140 $\pm$ 0.00001	0.0126$\pm$0.00001	1.11$\times$
sumObjIds	F	3/5	40%	0.0134 $\pm$ 0.00001	0.0105$\pm$0.00002	1.28$\times$
totalHeapSize	F	3/5	40%	0.0133 $\pm$ 0.00001	0.0102$\pm$0.00002	1.30$\times$
sweepUnmarked	M	5/5	0%	0.0580 $\pm$ 0.00011	0.0303$\pm$0.00008	1.92$\times$
touchHotObjects	M	5/5	0%	0.0572 $\pm$ 0.00016	0.0320$\pm$0.00001	1.79$\times$
Total				0.2121	0.1428	1.49$\times$
Geomean						1.33$\times$

Table 13: Per-pass performance for `PiecewiseFunctions`, ADT has 8 fields, SoA uses 7 buffers. Times are median per iteration (s); $\pm$ shows standard error of the mean across $-iterate$ runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	AoS med $\pm$ err	SoA med $\pm$ err	Speedup
autorefineMaxLevel	F	4/8	50%	0.0229 $\pm$ 0.00001	0.0197$\pm$0.00002	1.16$\times$
compressMass	F	3/8	62%	0.0207 $\pm$ 0.00001	0.0117$\pm$0.00001	1.77$\times$
lbDeuxLoadProxy	F	5/8	38%	0.0229 $\pm$ 0.00053	0.0185$\pm$0.00002	1.24$\times$
norm2Estimate	F	5/8	38%	0.0261 $\pm$ 0.00001	0.0179$\pm$0.00003	1.45$\times$
pmapCutHistogram	F	4/8	50%	0.0330 $\pm$ 0.00002	0.0128$\pm$0.00001	2.58$\times$
truncateTolViolations	F	3/8	62%	0.0211 $\pm$ 0.00001	0.0116$\pm$0.00068	1.82$\times$
addConstPW	M	8/8	0%	0.1026 $\pm$ 0.00025	0.0156$\pm$0.00004	6.55$\times$
diffPW	M	8/8	0%	0.1023 $\pm$ 0.00022	0.0310$\pm$0.00002	3.30$\times$
Total				0.3515	0.1388	2.53$\times$
Geomean						2.10$\times$

Table 14: Per-pass performance for `TernaryTree`, ADT has 5 fields, SoA uses 3 buffers. Times are median per iteration (s); $\pm$ shows standard error of the mean across $-iterate$ runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	AoS med $\pm$ err	SoA med $\pm$ err	Speedup
sum tree	F	5/5	0%	0.0285 $\pm$ 0.00007	0.0289 $\pm$ 0.00002	0.98 $\times$
add 1 tree	M	5/5	0%	0.0524 $\pm$ 0.00098	0.0408$\pm$0.00010	1.29$\times$
Total				0.0809	0.0697	1.16$\times$
Geomean						1.13$\times$

Table 15: Per-pass performance for `Trie`, ADT has 10 fields, SoA uses 9 buffers. Times are median per iteration (s); $\pm$ shows standard error of the mean across $-iterate$ runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	AoS med $\pm$ err	SoA med $\pm$ err	Speedup
autocompleteTopKProxy	F	5/10	50%	0.0158 $\pm$ 0.00001	0.0080$\pm$0.00003	1.98$\times$
countLazyNodes	F	4/10	60%	0.0136 $\pm$ 0.00000	0.0065$\pm$0.00001	2.10$\times$
countTerminals	F	3/10	70%	0.0129 $\pm$ 0.00040	0.0058$\pm$0.00000	2.24$\times$
sumPrefixFreq	F	3/10	70%	0.0130 $\pm$ 0.00008	0.0056$\pm$0.00001	2.31$\times$
sumSubtreeHints	F	3/10	70%	0.0130 $\pm$ 0.00001	0.0057$\pm$0.00001	2.29$\times$
decayTrieStats	M	10/10	0%	0.0675 $\pm$ 0.00023	0.0242$\pm$0.00007	2.79$\times$
resetTraversalState	M	10/10	0%	0.0634 $\pm$ 0.00012	0.0150$\pm$0.00001	4.21$\times$
Total				0.1992	0.0707	2.82$\times$
Geomean						2.48$\times$